

CEWICE LEBUNIA, Poland

WMO: 121360

Lat: 54.417N

Lon: 17.767E

Elev: 151

StdP: 99.52

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.2	-11.9	-18.0	0.8	-14.2	-14.8	1.0	-11.3	12.4	4.2	10.9	4.3	1.5	110	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	28.8	19.0	26.9	18.4	25.0	17.5	20.6	26.1	19.6	24.6	18.5	23.1	3.8	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	13.9	22.9	17.9	13.1	21.5	16.9	12.2	20.7	60.1	26.0	56.5	24.5	53.2	22.9	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.7	8.3	7.3	-20.2	32.2	5.5	2.3	-24.1	33.8	-27.3	35.2	-30.3	36.5	-34.3	38.1		
(5)			-20.0	22.6	5.4	1.1	-23.9	23.4	-27.1	24.0	-30.1	24.7	-34.0	25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.8	-2.0	-1.0	2.1	7.1	11.5	15.0	17.5	17.2	13.1	8.1	4.1	0.6
	DBStd	7.79	5.12	4.63	4.32	4.01	4.16	3.36	3.15	2.95	3.12	3.52	3.25	4.52
	HDD10.0	1616	372	309	246	104	30	2	0	0	7	79	177	291
	HDD18.3	3904	630	542	503	337	216	111	56	57	161	316	427	549
	CDD10.0	822	0	0	1	18	77	151	231	224	98	21	1	0
	CDD18.3	68	0	0	0	0	4	11	28	23	2	0	0	0
	CDH23.3	798	0	0	0	8	64	151	321	227	27	0	0	0
	CDH26.7	184	0	0	0	1	8	37	90	44	4	0	0	0
Wind	WSAvg	3.3	3.7	3.6	3.6	3.4	3.2	3.2	3.0	2.8	2.9	3.3	3.4	3.7
Precipitation	PrecAvg	729	54	41	48	35	63	74	89	77	70	61	61	61
	PrecMax	963	162	81	120	79	120	178	214	130	154	155	135	127
	PrecMin	540	4	7	12	2	16	4	15	31	26	18	10	17
	PrecStd	114	34	18	27	20	26	39	45	25	34	37	30	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	9.2	16.0	24.1	27.7	30.8	32.0	30.1	26.7	19.8	12.7	10.2
		MCWB	8.1	7.0	9.5	13.8	17.3	20.7	20.2	20.6	17.9	14.4	9.8	9.0
	2%	DB	7.1	7.5	13.1	20.4	25.1	27.1	29.1	27.8	23.2	17.1	10.9	8.8
		MCWB	6.1	5.6	7.7	12.1	16.6	18.3	19.0	19.2	16.3	13.2	9.5	8.1
	5%	DB	5.8	5.9	10.6	17.9	22.8	24.8	26.8	26.0	21.1	15.2	9.9	7.2
		MCWB	5.0	4.5	6.9	10.5	15.2	16.8	18.7	18.4	15.4	12.3	8.7	6.3
	10%	DB	4.3	4.7	8.2	15.5	20.3	22.5	24.8	24.0	19.1	13.7	8.8	6.1
		MCWB	3.7	3.5	5.8	9.3	13.7	15.7	17.9	17.8	14.6	11.5	7.9	5.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.2	8.1	10.7	15.2	19.1	21.6	22.0	21.9	19.1	15.4	10.7	9.5
		MCDB	9.1	8.8	14.2	24.0	24.3	27.7	28.5	28.1	23.9	18.1	11.6	10.0
	2%	WB	6.3	5.7	8.7	12.7	17.5	19.4	20.7	20.3	17.4	14.0	9.6	7.9
		MCDB	6.8	6.8	11.2	18.7	24.0	24.9	26.2	25.3	21.5	16.4	10.5	8.5
	5%	WB	4.8	4.5	7.3	11.2	16.0	17.8	19.7	19.3	16.2	12.8	8.8	6.5
		MCDB	5.4	5.5	9.8	16.8	21.4	23.1	24.9	24.0	19.8	14.8	9.6	7.0
	10%	WB	3.7	3.4	5.9	9.8	14.5	16.5	18.7	18.3	15.2	11.7	7.8	5.3
		MCDB	4.3	4.4	7.9	14.3	19.3	21.1	22.9	22.5	18.4	13.4	8.6	5.9
Mean Daily Temperature Range	5% DB	MDBR	4.9	6.1	8.5	12.2	12.2	11.6	10.8	11.0	10.0	7.7	4.8	4.3
		MCDBR	4.3	7.1	12.5	16.8	15.6	15.7	14.9	14.5	13.3	10.1	5.9	4.4
		MCWBR	4.0	5.6	8.0	9.3	8.2	7.9	6.7	7.2	7.2	6.8	4.8	4.2
	5% WB	MCDBR	4.0	6.2	10.7	14.9	13.9	13.7	12.3	12.7	11.4	8.8	5.7	4.3
		MCWBR	3.9	5.3	7.6	8.6	7.6	7.4	6.3	7.0	6.7	6.3	4.8	4.2
Clear-Sky Solar Irradiance	taub	0.305	0.333	0.361	0.403	0.396	0.391	0.419	0.413	0.373	0.355	0.338	0.302	
	taud	2.270	2.275	2.276	2.221	2.297	2.340	2.281	2.301	2.393	2.416	2.326	2.273	
	Ebn at Noon	633	744	808	820	849	857	822	802	791	716	588	552	
	Edh at Noon	59	82	102	124	121	117	122	113	91	72	57	48	
All-Sky Solar Radiation	RadAvg	0.47	1.13	2.36	4.14	5.37	5.67	5.13	4.31	3.08	1.57	0.61	0.34	
	RadStd	0.06	0.16	0.25	0.49	0.58	0.44	0.58	0.41	0.33	0.23	0.09	0.04	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (27 neighbors)		+0.39	N/A	N/A	N/A	N/A	N/A	-130	+67	N/A

Nomenclature: See separate page